US 20250276707A1

(19) United States

(12) Patent Application Publication

(10) Pub. No.: US 2025/0276707 A1

EASWARAN

(43) Pub. Date: Sep. 4, 2025

(54) SYSTEM SAFETY ASSOCIATED WITH VEHICLE AUTONOMY

(71) Applicant: **Rivian IP Holdings, LLC**, Irvine, CA (US)

(72) Inventor: **Vasant Kumar EASWARAN**, Frisco, TX (US)

(21) Appl. No.: **19/053,214**

(22) Filed: **Feb. 13, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/560,591, filed on Mar. 1, 2024.

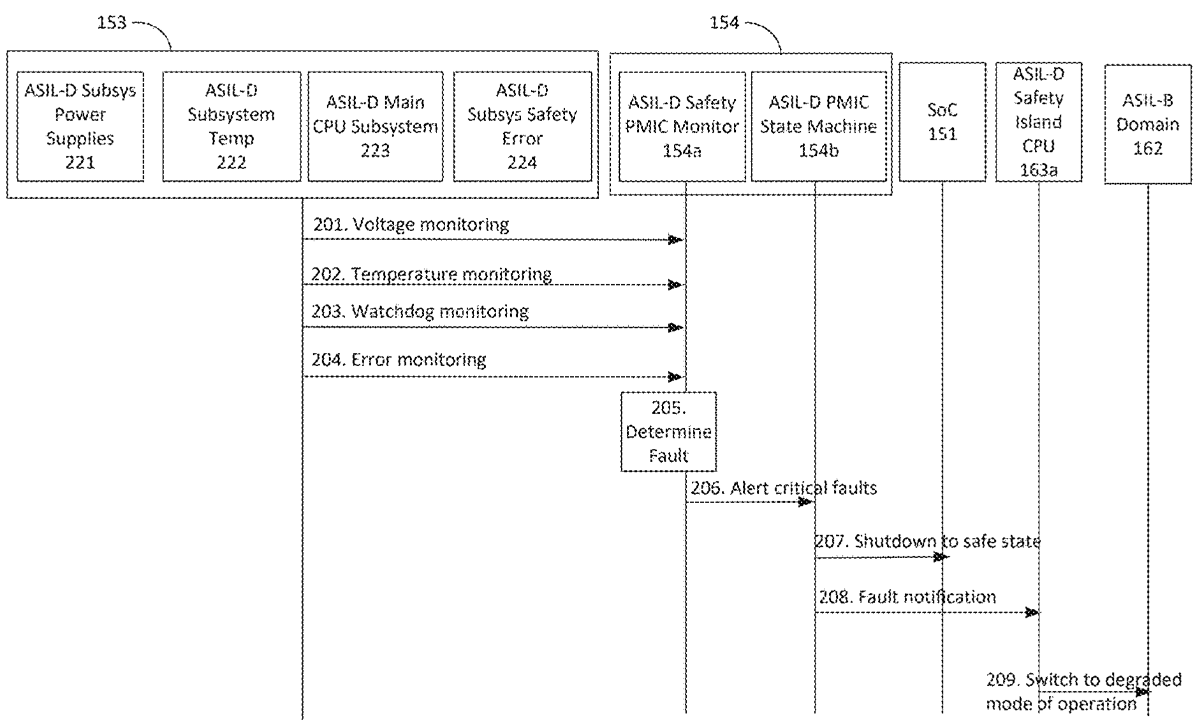
Publication Classification

(51) **Int. Cl.**
B60W 50/029 (2012.01)
B60W 50/02 (2012.01)
B60W 60/00 (2020.01)

(52) **U.S. Cl.**
CPC ***B60W 50/029*** (2013.01); ***B60W 50/0205*** (2013.01); ***B60W 60/00186*** (2020.02); *B60W 2050/0295* (2013.01); *B60W 2510/244* (2013.01); *B60W 2510/246* (2013.01)

(57) **ABSTRACT**

Methods, systems, and apparatuses, among other things, as described herein may provide for using intelligent power supplies or redundant SOCs to meet safety integrity or availability for Level 2 or Level 3 autonomy systems.



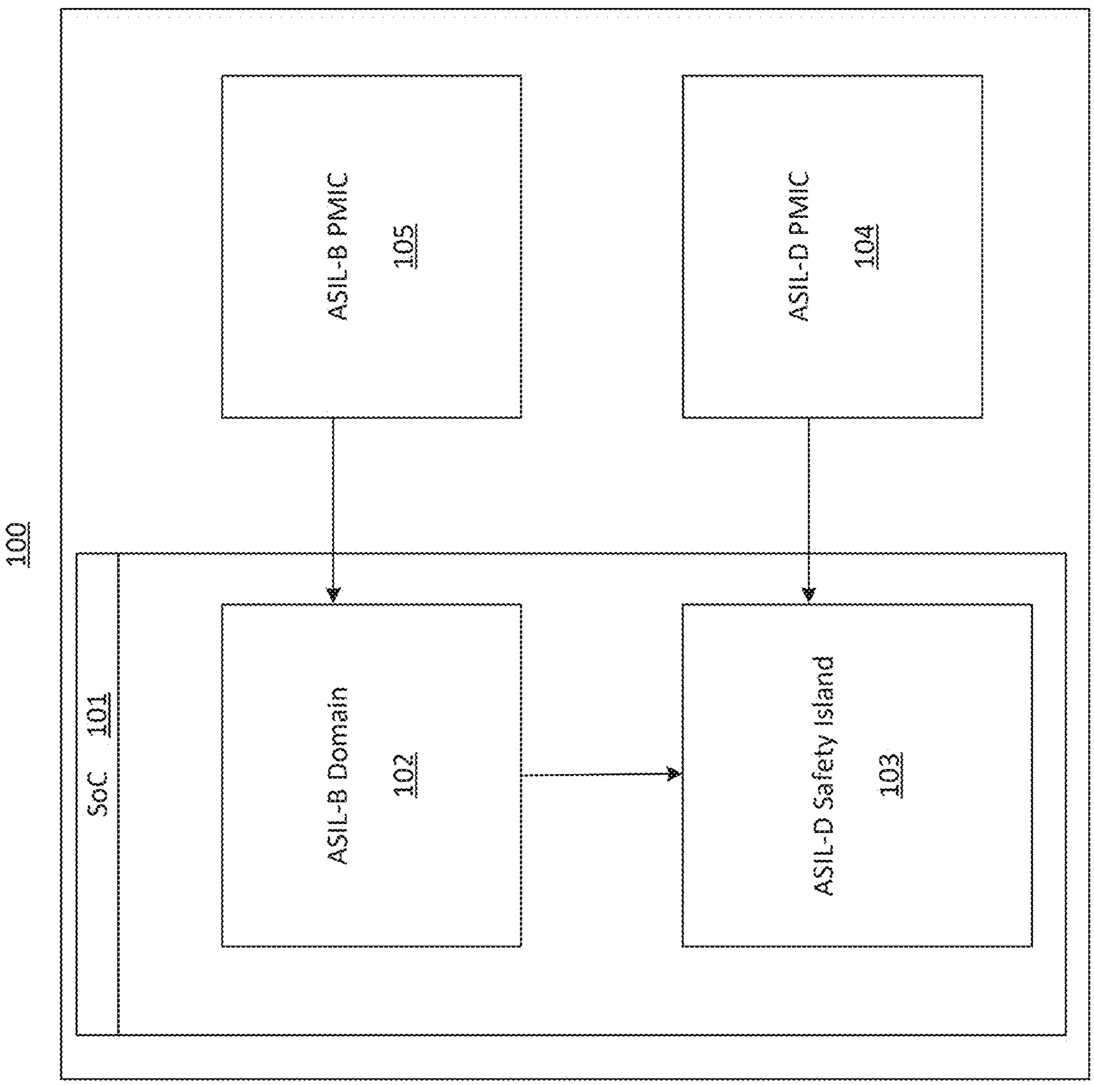

100
SoC 101
ASIL-B Domain
102
ASIL-D Safety Island
103
ASIL-B PMIC
105
ASIL-D PMIC
104

FIG. 1

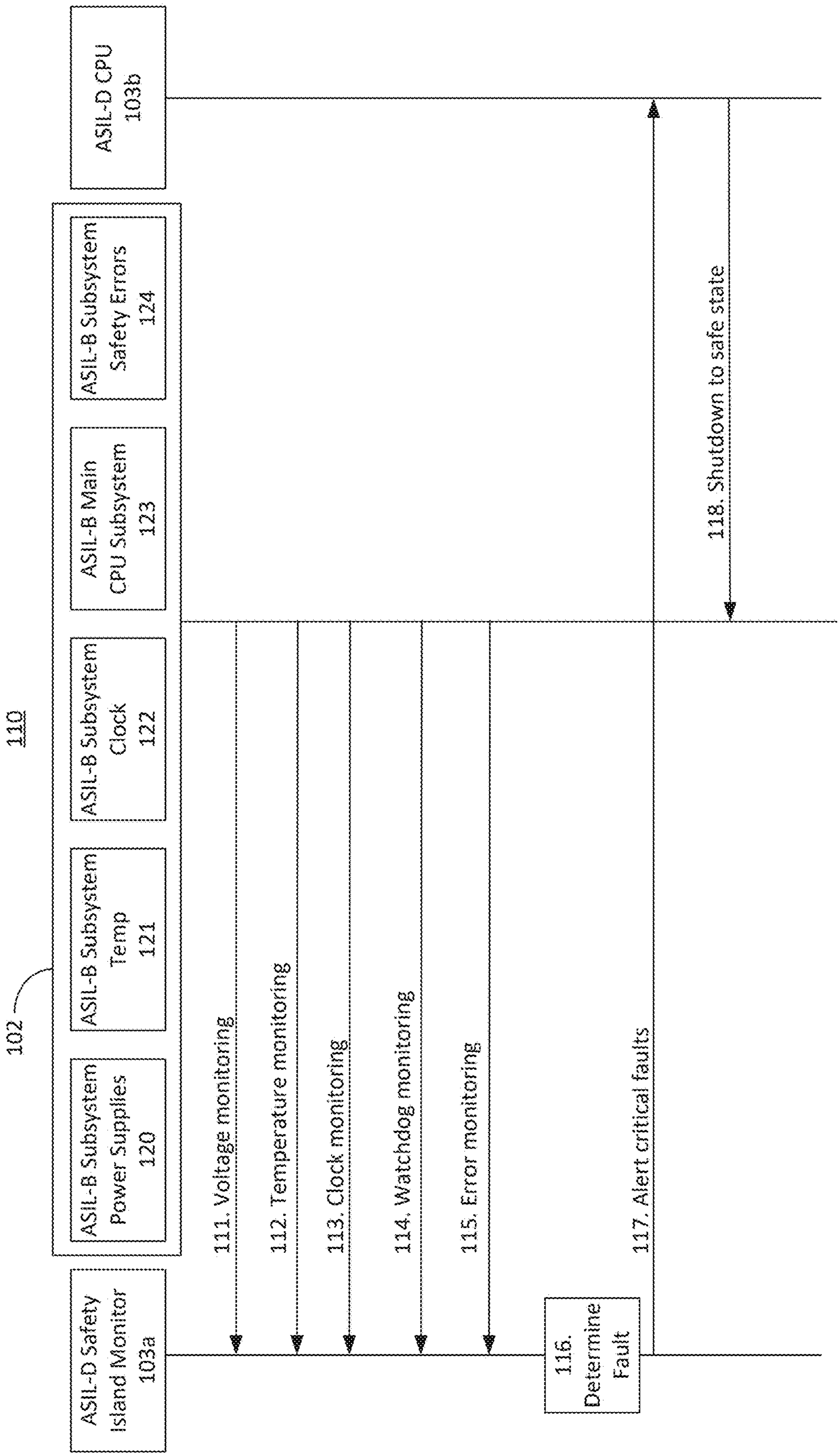

110
102
ASIL-D Safety Island Monitor 103a
ASIL-B Subsystem Power Supplies 120
ASIL-B Subsystem Temp 121
ASIL-B Subsystem Clock 122
ASIL-B Main CPU Subsystem 123
ASIL-B Subsystem Safety Errors 124
ASIL-D CPU 103b
111. Voltage monitoring
112. Temperature monitoring
113. Clock monitoring
114. Watchdog monitoring
115. Error monitoring
116. Determine Fault
117. Alert critical faults
118. Shutdown to safe state

FIG. 2

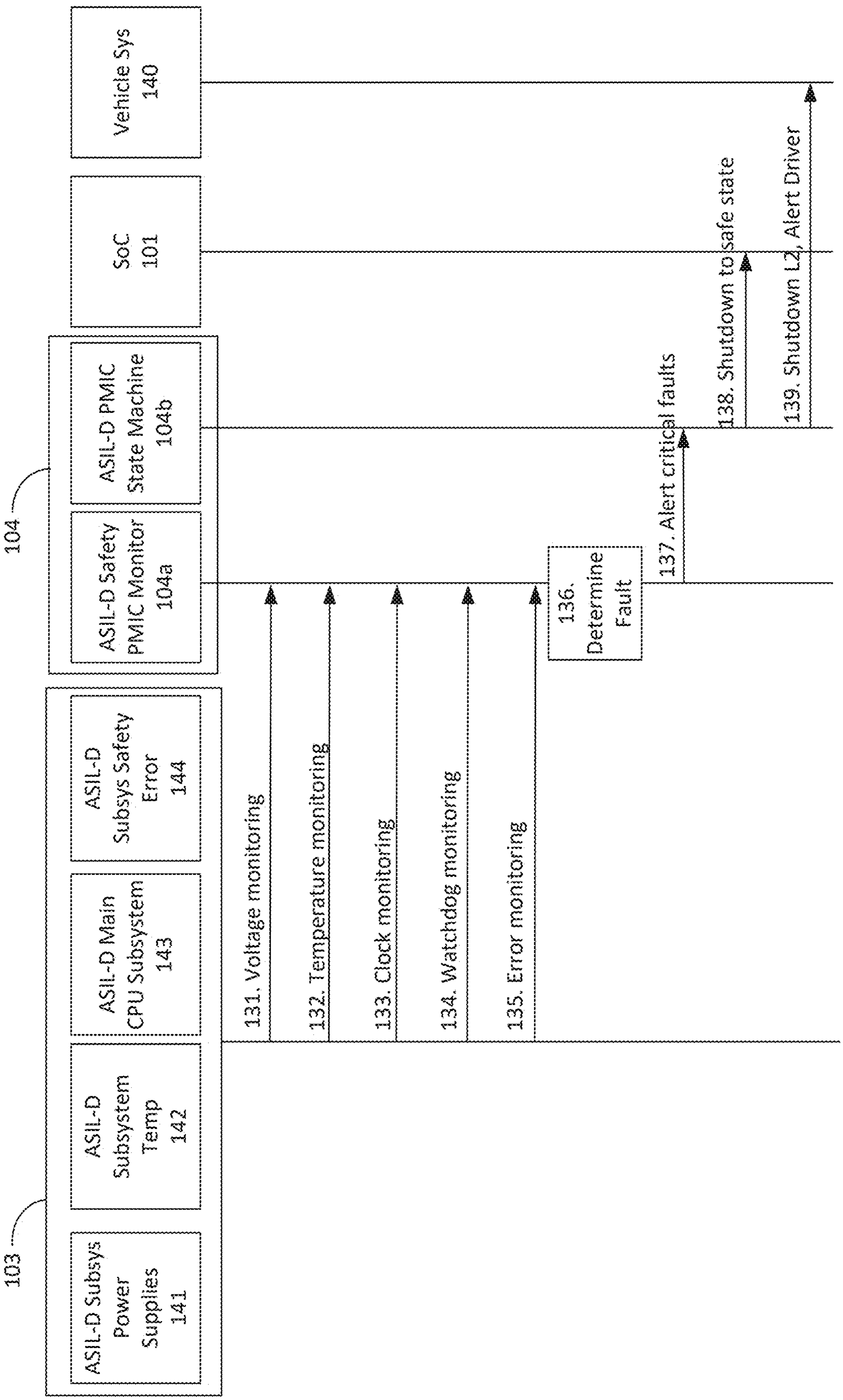

Vehicle Sys 140
SoC 101
104
ASIL-D PMIC State Machine 104b
ASIL-D Safety PMIC Monitor 104a
103
ASIL-D Subsys Safety Error 144
ASIL-D Main CPU Subsystem 143
ASIL-D Subsystem Temp 142
ASIL-D Subsys Power Supplies 141
131. Voltage monitoring
132. Temperature monitoring
133. Clock monitoring
134. Watchdog monitoring
135. Error monitoring
136. Determine Fault
137. Alert critical faults
138. Shutdown to safe state
139. Shutdown L2, Alert Driver

FIG. 3

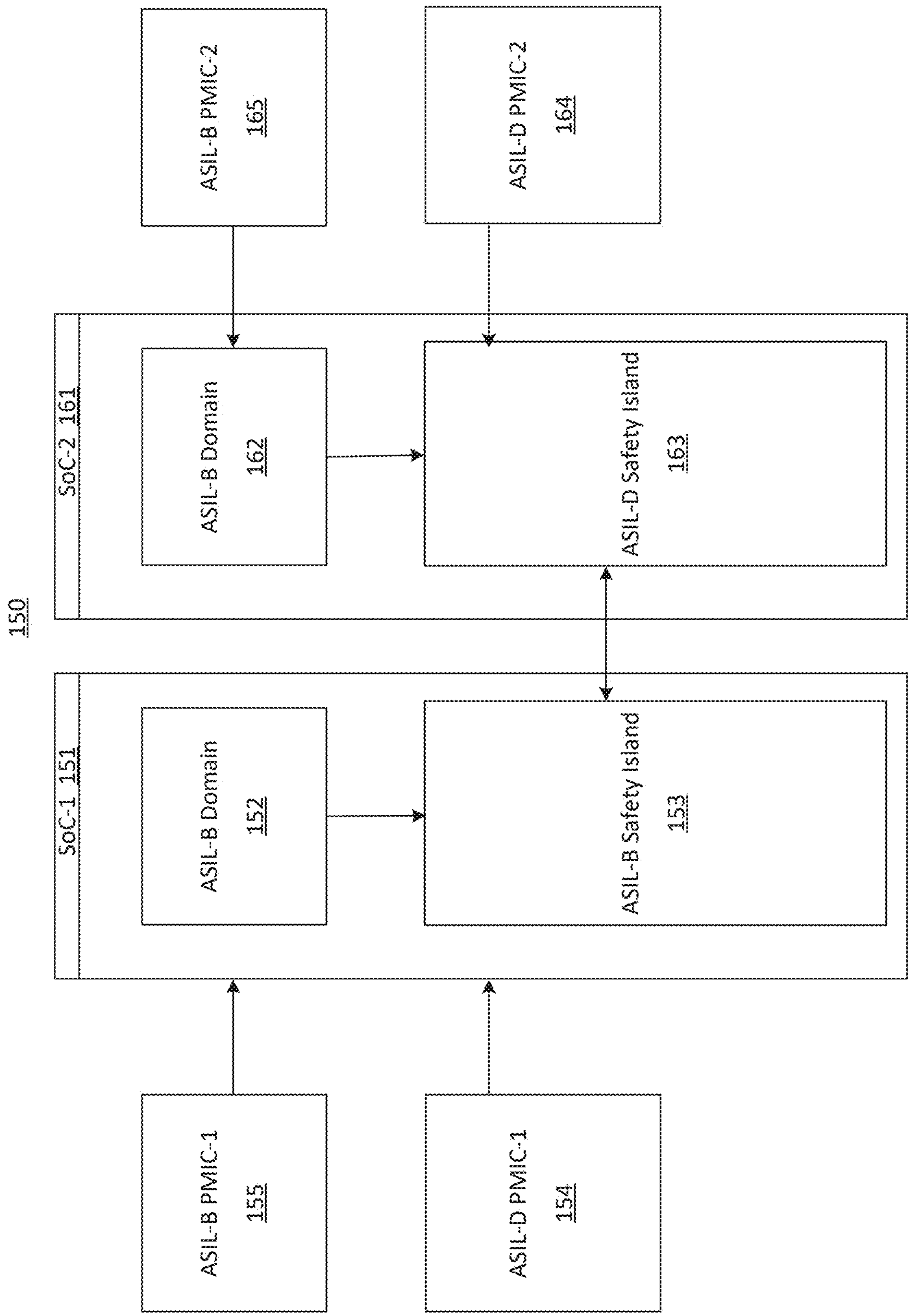

150
SoC-1 151
ASIL-B Domain
152
ASIL-B Safety Island
153
SoC-2 161
ASIL-B Domain
162
ASIL-D Safety Island
163
ASIL-B PMIC-1
155
ASIL-D PMIC-1
154
ASIL-B PMIC-2
165
ASIL-D PMIC-2
164

FIG. 4

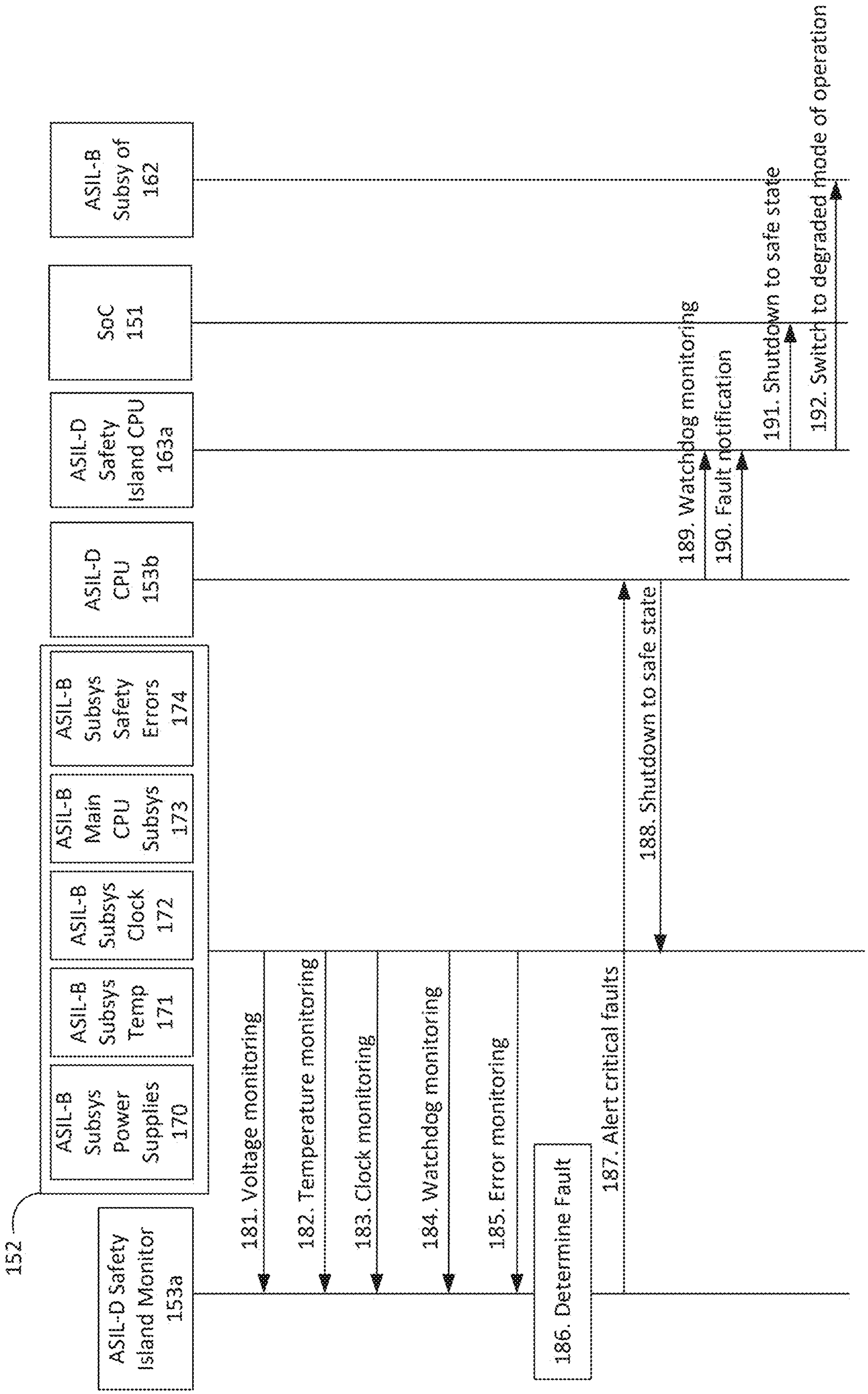

152
ASIL-D Safety Island Monitor 153a
ASIL-B Subsys Power Supplies 170
ASIL-B Subsys Temp 171
ASIL-B Subsys Clock 172
ASIL-B Main CPU Subsys 173
ASIL-B Subsys Safety Errors 174
ASIL-D CPU 153b
ASIL-D Safety Island CPU 163a
SoC 151
ASIL-B Subsy of 162
181. Voltage monitoring
182. Temperature monitoring
183. Clock monitoring
184. Watchdog monitoring
185. Error monitoring
186. Determine Fault
187. Alert critical faults
188. Shutdown to safe state
189. Watchdog monitoring
190. Fault notification
191. Shutdown to safe state
192. Switch to degraded mode of operation

FIG. 5

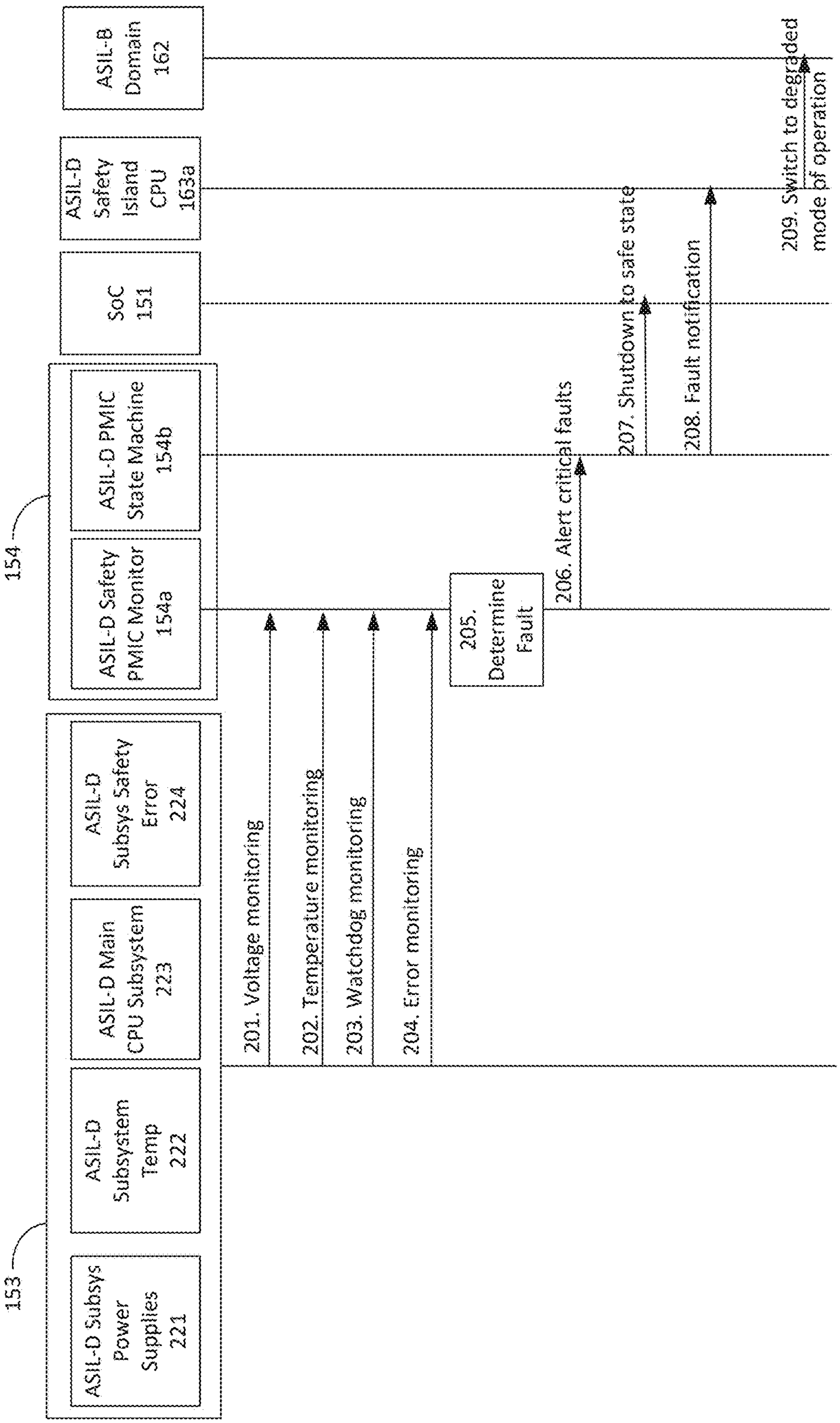

153
ASIL-D Subsys Power Supplies 221
ASIL-D Subsystem Temp 222
ASIL-D Main CPU Subsystem 223
ASIL-D Subsys Safety Error 224
154
ASIL-D Safety PMIC Monitor 154a
ASIL-D PMIC State Machine 154b
SoC 151
ASIL-D Safety Island CPU 163a
ASIL-B Domain 162
201. Voltage monitoring
202. Temperature monitoring
203. Watchdog monitoring
204. Error monitoring
205. Determine Fault
206. Alert critical faults
207. Shutdown to safe state
208. Fault notification
209. Switch to degraded mode of operation

FIG. 6

SYSTEM SAFETY ASSOCIATED WITH VEHICLE AUTONOMY

CROSS REFERENCE TO RELATED APPLICATION(S)

[0001] The present application claims the benefit of U.S. Provisional Application No. 63/560,591 entitled, "SYSTEM SAFETY FOR LEVEL 2 OR LEVEL 3 AUTONOMY", filed Mar. 1, 2024, the entirety of which is incorporated herein for reference.

INTRODUCTION

[0002] Automotive Safety Integrity Level (ASIL) is a risk classification system defined by the ISO 26262 standard for the functional safety of road vehicles. ASIL classifies hazards in one of four levels, denoted as A through D, with a fifth additional level for non-hazardous systems or components. ASIL D represents the highest level of risk, while ASIL A represents the lowest risk level.

[0003] The standard defines functional safety as "the absence of unreasonable risk due to hazards caused by malfunctioning behavior of electrical or electronic systems." ASILs establish safety requirements—based on the probability and acceptability of harm—for automotive components to be compliant with ISO 26262.

[0004] Systems that include vehicle brakes may require an ASIL-D grade—the highest rigor applied to safety assurance—because of the significant risks associated with their failure. ASIL-B examples are headlights and brake lights, while ASIL C may be for systems that include cruise control. Rear lights are example lights that may be classified with an ASIL-A grade.

[0005] Aspects of the subject technology can help to improve the overall cost, reliability, and efficiency of circuits or other electronic components.

SUMMARY

[0006] The present description is generally directed to intelligent power supplies or redundant SOCs that may be implemented to meet safety integrity and availability for Level 2 or Level 3 autonomy systems.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Certain features of the subject technology are set forth in the appended claims. However, for purpose of explanation, several embodiments of the subject technology are set forth in the following figures.

[0008] FIG. **1** illustrates an exemplary integrated circuit associated with safety or availability of Level 2 autonomy vehicles.

[0009] FIG. **2** illustrates an exemplary sequence diagram associated with a safety island monitoring a system-on-a-chip (SOC) in a Level 2 autonomy system.

[0010] FIG. **3** illustrates an exemplary sequence diagram associated with safety ASIL-D power management integrated circuit (PMIC) monitoring ASIL-D safety island and notifying the vehicle system in a Level 2 autonomy system.

[0011] FIG. **4** illustrates an exemplary integrated circuit associated with safety or availability of Level 3 autonomy vehicles.

[0012] FIG. **5** illustrates an exemplary sequence diagram associated with a seconds SOC and first safety island monitoring a first SOC in a Level 3 autonomy system.

[0013] FIG. **6** illustrates an exemplary sequence diagram associated with a second SOC and first safety ASIL-D PMIC monitoring a first ASIL-D safety island of a first SOC in a Level 3 autonomy system.

DETAILED DESCRIPTION

[0014] The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a thorough understanding of the subject technology. However, the subject technology is not limited to the specific details set forth herein and can be practiced using one or more other implementations. In one or more implementations, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

[0015] There are six levels of driving automation ranging from 0 (fully manual) to 5 (fully autonomous) that have been adopted by the U.S. Department of Transportation. In an example, with regard to Level 2 autonomy, there are advanced driver assistance systems (ADAS) that may provide continuous assistance with acceleration, braking, and steering, while a driver sits in the driver seat of a vehicle and can take control of the vehicle at any time. The driver remains engaged and attentive. ASIL-rated SoCs may be used to implement such automated driving.

[0016] Vehicles executing Level 3 autonomy may have environmental detection capabilities and can make informed decisions for themselves, such as accelerating past a slow-moving vehicle. The Level 3 system actively performs driving tasks while a driver remains available to take over. If the system can no longer operate and prompts the driver, the driver should be available to resume driving tasks.

[0017] ASIL rated SoCs typically have several safety critical use cases which are required to meet differing levels of safety integrity. Due to the differing coverage and residual failure rates of different ASIL levels, care is taken that the requirements are met with mitigations planned for any interference between the ASIL domains. To address monitoring needs of vehicle systems (e.g., one or more electronic components), the safety goals may be realized via various levels of ASIL decomposition. While doing these decompositions, there may be safe isolations planned at the intersection of boundaries for electrical, data, or control paths of vehicle systems that may allow for safe detection of faults (e.g., errors) of electronic components.

[0018] The isolation capability may ensure availability of higher integrity functions in the event of faults detected in lower integrity functions. In implementing this isolation capability, the availability requirements in Level 2 autonomy systems may be addressed opportunistically since the fail-over is back to a driver in such systems. As discussed in more detail herein, intelligent power supplies—instead of high integrity micro controllers—may be implemented to meet safety integrity and availability for Level 2 autonomy systems at a possibly lower cost point.

[0019] FIG. **1** illustrates an exemplary integrated circuit associated with safety or availability of Level 2 autonomy vehicles. With regard to Level 2 autonomy, system **100** may include system-on-a-chip (SOC) **101**. SOC **101** may include

ASIL-B domain **102** (e.g., a power domain) or ASIL-D safety island **103**. ASIL-B domain **102** may include safety mechanisms that achieve ASIL-B integrity across safety critical use cases that use different subsystems in this ASIL-B domain **102**. ASIL-B domain **102** may be communicatively connected with one or monitor ASIL-B power management integrated circuit (PMIC) **105**, ASIL-D safety island **103**, or ASIL-D PMIC **104**. ASIL-B PMIC **105** may include safety mechanisms that achieve ASIL-B integrity across safety critical voltage rails powering the ASIL-B SoC domain (e.g., ASIL-B domain **102**).

[0020] With continued reference to FIG. **1**, ASIL-D safety island **103** may include processors with hardware features, such as error correction code (ECC) and a programmable watchdog timer to detect system failures or runtime faults. The processors may include a lockstep interface that is used by the integrated safety monitor to compare outputs and detect if a fault has occurred. ASIL-D safety island **103** may include safety mechanisms that achieve ASIL-D integrity across safety critical use cases that use different subsystems in this domain. In addition, safety mechanisms of ASIL-D safety island **103** may monitor ASIL-B domains (e.g., ASIL-B domain **102**) to achieve higher levels of integrity through ASIL decomposition.

[0021] ASIL-D PMIC **104** may include safety mechanisms that achieve ASIL-D integrity across safety critical voltage rails powering ASIL-D safety island **103**. In addition, safety mechanisms of ASIL-D PMIC **104** may monitor ASIL-D safety island **103** to achieve higher levels of integrity by mitigating risk of dependent failures within SOC **101**. Instead of using a microcontroller (MCU), ASIL-D PMIC **104** may be used to monitor SOC **101**, as shown in FIG. **1**. SOC **101** may include ASIL-D safety island **103** and ASIL-B domain **102**. The minimal functions that may be provided by the MCU may be implemented by ASIL-D PMIC **104**. The use of ASIL-D PMIC **104** as disclosed is sufficient for the required safety integrity. ASIL-D PMIC **104** may not only include monitoring capability for power rails, but other monitoring capabilities, such as a watchdog timer. In an example, SOC **101**, through its communications interfaces within the integrated circuit may write to certain registers in the watchdog timer at a certain frequency (e.g., 50 ms periodically). When the frequency of writing to the register is maintained, the watchdog timer may reset its counter every time. When the frequency of writing to the register is not maintained, an alert may be indicated and ASIL-D PMIC **104** (since it controls power management) may shut off the power rails to the SOC **101**, which may take the vehicle system to a safe state. And through the communication interfaces of ASIL-D PMIC **104**, higher level systems may be alerted of the fault, which may eventually be displayed (e.g., display a message that level 2 autonomy has failed).

[0022] FIG. **2** illustrates an exemplary sequence diagram of system **110** associated with safety island **103** monitoring SOC **101**. At step **111** through step **115**, ASIL-D safety island monitor **103***a* may monitor the state of one or more ASIL-B subsystems. ASIL-B subsystems of ASIL-B domain **102** may include one or more ASIL-B subsystems **120** for power supplies (e.g., associated with step **111**), ASIL-B subsystems **121** for temperatures (e.g., associated with step **112**), ASIL-B subsystems **122** for clock (e.g., associated with step **113**), ASIL-B subsystems **123** for main CPU (e.g., associated with step **114**), or ASIL-B subsystems **124** for safety errors (e.g., associated with step **115**), among others.

[0023] The monitoring may be based on receiving information within a designated threshold or not receiving information within a designated period. At step **116**, based on the monitoring of step **111**-step **115**, a fault may be determined. At step **117**, for example, an alert associated with a critical fault may be sent to ASIL-D CPU **103***b*. At step **118**, based on receiving the alert, ASIL-D CPU **103***b* may send indication to one or more ASIL-B subsystems of ASIL-B domain **102**, to shutdown, which may place a vehicle system into a safe state. It is contemplated herein that ASIL-D safety island **103** may include ASIL-D safety island monitor **103***a* or ASIL-D CPU **103***b*.

[0024] FIG. **3** illustrates an exemplary sequence diagram associated with safety ASIL-D PMIC **104** monitoring ASIL-D safety island **103** and notifying the vehicle system **140**. At step **131** through step **135**, ASIL-D PMIC **104** (e.g., ASIL-D safety PMIC monitor **104***a*) may monitor the state of one or more ASIL-D safety island **103** subsystems. ASIL-D safety island **103** subsystems may include one or more ASIL-D subsystems **141** for power supplies (e.g., associated with step **131**), ASIL-D subsystems **142** for temperatures (e.g., associated with step **132**), ASIL-D subsystems **143** for main CPU (e.g., associated with step **134**), or ASIL-D subsystems **144** for safety errors (e.g., associated with step **135**), among others (e.g., step **133**).

[0025] The monitoring may be based on receiving information within a designated threshold or not receiving information within a designated period. At step **136**, based on the monitoring of step **131**-step **135**, a fault may be determined. At step **137**, for example, an alert associated with a critical fault may be sent to ASIL-D PMIC state machine **104***b*. At step **138**, based on receiving the alert, ASIL-D PMIC state machine **104***b* may send an indication to one or more components of SOC **101**, to shutdown, which may place a vehicle system into a safe state. At step **139**, ASIL-D PMIC state machine **104***b* may shutdown Level 2 autonomy and send an alert for the driver. It is contemplated that the components of processing flow of FIG. **2** or FIG. **3** may be on a single component or may be functional components. Vehicle system **140** may include one or more components of FIG. **1**.

[0026] FIG. **4** illustrates an exemplary integrated circuit associated with safety or availability of automated vehicles. With regard to Level 3 autonomy, system **150** may include system-on-a-chip (SOC) **151**. SOC **151** may include ASIL-B domain **152** (e.g., a power domain) or ASIL-D safety island **153**. ASIL-B domain **152** may include safety mechanisms that achieve ASIL-B integrity across safety critical use cases that use different subsystems in this ASIL-B domain **152**. ASIL-B domain **152** may be communicatively connected with one or monitor ASIL-B PMIC **155**, ASIL-D safety island **153**, or ASIL-D PMIC **154**. ASIL-B PMIC **155** may include safety mechanisms that achieve ASIL-B integrity across safety critical voltage rails powering the ASIL-B SoC domain (e.g., ASIL-B domain **152**). SOC **151** may be communicatively connected with SOC **161**, which may be connected via ASIL-D safety island **153** with ASIL-D safety island **163**.

[0027] With continued reference to FIG. **4**, ASIL-D safety island **153** may include safety mechanisms that achieve ASIL-D integrity across safety critical use cases that use different subsystems in this domain. Safety mechanisms of

ASIL-D safety island **153** may monitor ASIL-B domains (e.g., ASIL-B domain **102**) to achieve higher levels of integrity through ASIL decomposition. In an event of an irrecoverable fault in this ASIL-D domain (e.g., ASIL-D safety island **153**), an external ASIL-D domain (e.g., ASIL-D safety island **163**) may monitor ASIL-D safety island **153** and initiate a degraded mode of operation in the event of a fault. Similarly, this ASIL-D domain (e.g., ASIL-D safety island **153**) may also monitor an external ASIL-D domain (e.g., ASIL-D safety island **163**) and initiate a degraded mode of operation in the event of a fault.

[0028] ASIL-D PMIC **154** may include safety mechanisms that achieve ASIL-D integrity across safety critical voltage rails powering ASIL-D safety island **153**. In addition, safety mechanisms of ASIL-D PMIC **154** may monitor ASIL-D safety island **153** to achieve higher levels of integrity by mitigating risk of dependent failures within the SOC **151**.

[0029] With continued reference to FIG. **4**, system **150** may include system-on-a-chip (SOC) **161**. SOC **101** may include ASIL-B domain **162** (e.g., a power domain) or ASIL-D safety island **163**. ASIL-B domain **162** may include safety mechanisms that achieve ASIL-B integrity across safety critical use cases that use different subsystems in this ASIL-B domain **162**. ASIL-B domain **162** may be communicatively connected with one or monitor ASIL-B PMIC **165**, ASIL-D safety island **163**, or ASIL-D PMIC **164**. ASIL-B PMIC **165** may include safety mechanisms that achieve ASIL-B integrity across safety critical voltage rails powering the ASIL-B SoC domain (e.g., ASIL-B domain **162**). SOC **161** may be communicatively connected with SOC **161**, which may be connected via ASIL-D safety island **163** with ASIL-D safety island **153**.

[0030] With continued reference to FIG. **4**, ASIL-D safety island **163** may include safety mechanisms that achieve ASIL-D integrity across safety critical use cases that use different subsystems in this domain. Safety mechanisms of ASIL-D safety island **163** may monitor ASIL-B domains (e.g., ASIL-B domain **162**) to achieve higher levels of integrity through ASIL decomposition. In an event of an irrecoverable fault in this ASIL-D domain (e.g., ASIL-D safety island **163**), an external ASIL-D domain (e.g., ASIL-D safety island **153**) may be monitoring this domain and initiate a degraded mode of operation in the event of a fault. Similarly, this ASIL-D domain (e.g., ASIL-D safety island **163**) may also monitor an external ASIL-D domain (e.g., ASIL-D safety island **153**) and initiate a degraded mode of operation in the event of a fault.

[0031] ASIL-D PMIC **164** may include safety mechanisms that achieve ASIL-D integrity across safety critical voltage rails powering ASIL-D safety island **163**. In addition, safety mechanisms of ASIL-D PMIC **164** may monitor ASIL-D safety island **163** to achieve higher levels of integrity by mitigating risk of dependent failures within the SOC **161**. As disclosed, when the two SOCs (SOC **151** and SOC **161**) are communicating with each other, they can also monitor each other. Therefore, in an example, SOC **161** or ASIL-D PMIC **154** may monitor ASIL-D safety island **153** and SOC **151** or ASIL-D PMIC **164** may monitor ASIL-D safety island **163**. This particularly may be implemented for level 3 autonomy. This configuration may reduce the number of hops needed for alerting SOC **161** (if SOC **151** is in a fault condition) and trigger SOC **161** to implement a degraded functionality. For example, instead of two hops (e.g., ASIL-D safety island **153** to ASIL-D PMIC **154** to SOC **161**) there may be one hop (e.g., ASIL-D safety island **153** to SOC **161**). Degraded functionality (also referred herein as degraded mode of operation) may include reducing from Level 3 to Level 2, performing automated driving at a designated speed for a period (e.g., 40 mph for 30 seconds, 75% or less of the posted speed limit of the road, or the like), submitting control to a driver, or stopping the vehicle.

[0032] FIG. **5** illustrates an exemplary sequence diagram associated with SOC **161** and safety island **153** monitoring SOC **151**. At step **181** through step **185**, ASIL-D safety island monitor **153*a*** may monitor the state of one or more ASIL-B subsystems of ASIL-B domain **152**. ASIL-B subsystems of ASIL-B domain **152** may include one or more ASIL-B subsystems **170** for power supplies (e.g., associated with step **181**), ASIL-B subsystems **171** for temperatures (e.g., associated with step **182**), ASIL-B subsystems **172** for clock (e.g., associated with step **183**), ASIL-B subsystems **173** for main CPU (e.g., associated with step **184**), or ASIL-B subsystems **174** for safety errors (e.g., associated with step **185**), among others.

[0033] The monitoring may be based on receiving information within a designated threshold or not receiving information within a designated period. At step **186**, based on the monitoring of step **181**-step **185**, a fault may be determined. At step **187**, for example, an alert associated with a critical fault may be sent to ASIL-D CPU **153*b***. At step **188**, based on receiving the alert, ASIL-D CPU **153*b*** may send an indication to one or more ASIL-B subsystems of ASIL-B domain **152**, to shutdown, which may place a vehicle system into a safe state. At step **189**, ASIL-D safety island CPU **163*a*** may monitor ASIL-D CPU **153*b*** (e.g., via watchdog timer monitor). At step **190**, ASIL-D CPU **153*b*** may send a fault notification (which may be based on alert of step **187**) to ASIL-D safety island CPU **163*a***. The fault notification may be sent via error pins or serial peripheral (SPI) communication. At step **191**, ASIL-D safety island CPU **163*a*** may send, to SoC **151**, an indication to shutdown to get into a safe state. At step **192**, ASIL-D safety island CPU **163*a*** may send an indication to ASIL-B subsystem of ASIL-B domain **162** to switch to a degraded mode of operation. It is contemplated herein that ASIL-D safety island **153** may include ASIL-D safety island monitor **153*a*** or ASIL-D CPU **153*b***.

[0034] FIG. **6** illustrates an exemplary sequence diagram associated with SOC **161** and safety ASIL-D PMIC **154** monitoring ASIL-D safety island **153** of SOC **151**. At step **201** through step **204**, ASIL-D PMIC **154** (e.g., ASIL-D safety PMIC monitor **154*a***) may monitor the state of one or more ASIL-D safety island **153** subsystems. ASIL-D safety island **153** subsystems may include one or more ASIL-D subsystems **221** for power supplies (e.g., associated with step **201**), ASIL-D subsystems **222** for temperatures (e.g., associated with step **202**), ASIL-D subsystems **223** for main CPU (e.g., associated with step **203**), or ASIL-D subsystems **224** for safety errors (e.g., associated with step **204**), among others.

[0035] The monitoring may be based on receiving information within a designated threshold or not receiving information within a designated period. At step **205**, based on the monitoring of step **201**-step **204**, a fault may be determined. At step **206**, for example, an alert associated with a critical fault may be sent to ASIL-D PMIC state machine **154*b***. At step **207**, based on receiving the alert, ASIL-D PMIC state machine **154***b* may send an indication to one or more components of SOC **151** to shut down to get into a safe state. At step **208**, ASIL-D PMIC state machine **154***b* may send a fault notification, which may be via error pins, to one or more components of SOC **151**. At step **209**, ASIL-D Safety Island CPU **163***a* of SOC **161** may send to SOC **161** (e.g., ASIL-B domain **162** or ASIL-D safety island CPU **163***a*) an indication to switch to a degraded mode of operation (e.g., from Level 3 to Level 2 or driver takeover). It is contemplated that the components of processing flow of FIG. **5** or FIG. **6** may be on a single component or may be functional components. It is also contemplated that each SOC checks on each other, so the steps associated with FIG. **5** and FIG. **6** may be appropriately mirrored.

[0036] The disclosed subject matter herein associated with safety or availability of automated vehicles for Level 2 or Level 3 autonomy may be more frequently implemented in scenarios with vertical integration of a robust SOC **101**, in which one entity may be responsible for the design of the majority of the system. For example, if the SOC **101** is designed as disclosed then ASIL-D PMIC **104** may be sufficient instead of or in addition to an MCU, which may save cost and meet the required integrity. There may be multiple ways to achieve different levels of ASIL.

[0037] The disclosed subject matter may be used in or with automotive electronic components. Electronic components may be integrated into automobiles, such as an electric vehicle. The disclosed subject matter may result in a reduced number of components, reduced complexity of a system, or more efficient communication between components. In an exemplary implementation, the system proceeds to a safe state, such as shutting down an autonomous system application, which would functionally be safe.

[0038] Methods, systems, and apparatuses, among other things, as described herein may provide for using intelligent power supplies or redundant SOCs to meet safety integrity or availability for Level 2 or Level 3 autonomy systems. All combinations of the steps disclosed herein (including the removal or addition of steps or components) are contemplated in a manner that is consistent with the other portions of the detailed description.

[0039] Systems and methods for monitoring safety in autonomous vehicles are disclosed herein. A system may provide for an ASIL-B domain; an ASIL-D safety island communicatively connected with the ASIL-B domain; and an ASIL-D power management integrated circuit (PMIC) communicatively connected with the ASIL-D safety island; wherein the ASIL-D PMIC monitors operation of the ASIL-D safety island and initiates a shutdown sequence upon detecting a fault condition. The system may further include an ASIL-B PMIC communicatively connected with the ASIL-B domain, wherein the ASIL-B PMIC comprises safety mechanisms achieving ASIL-B integrity across safety critical voltage rails powering the ASIL-B domain. All combinations (including the removal or addition of steps) in this paragraph and the above paragraphs are contemplated in a manner that is consistent with the other portions of the detailed description.

[0040] A method for level two or level three autonomy may include monitoring, by an automotive safety integrity level-D (ASIL-D) power management integrated circuit (PMIC), operation of a first ASIL-D safety island domain of a first SOC, wherein the SOC comprises the first ASIL-D safety island domain communicatively connected with one or more ASIL-B subsystems of an ASIL-B domain; determining, by the ASIL-D PMIC, a fault condition associated with the first ASIL-D safety island domain; and based on determining that there is the fault, transmitting an indication to enter into a degraded mode of operation. The method may include monitoring, by a second ASIL-D safety island domain of a second SOC, the first ASIL-D safety island domain of the first SOC. The monitoring operation of the first ASIL-D safety island comprises monitoring one or more of power supplies, temperatures, or watchdog timers of the first ASIL-D safety island. Based on the indication of the fault condition, an indication to shutdown one or more components of the first SOC to a safe state may be transmitted. The degraded mode of operation comprises reducing from level three autonomy to level two autonomy, wherein the one or more applications comprise an autonomous driving application. All combinations (including the removal or addition of steps) in this paragraph and the above paragraphs are contemplated in a manner that is consistent with the other portions of the detailed description.

[0041] A reference to an element in the singular is not intended to mean one and only one unless specifically so stated, but rather one or more. For example, "a" module may refer to one or more modules. An element proceeded by "a," "an," "the," or "said" does not, without further constraints, preclude the existence of additional same elements.

[0042] Headings and subheadings, if any, are used for convenience only and do not limit the invention. The word exemplary is used to mean serving as an example or illustration. To the extent that the term include, have, or the like is used, such term is intended to be inclusive in a manner similar to the term comprise as comprise is interpreted when employed as a transitional word in a claim. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions.

[0043] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase (s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0044] A phrase "at least one of" preceding a series of items, with the terms "and" or "or" to separate any of the items, modifies the list as a whole, rather than each member of the list. The phrase "at least one of" does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, each of the phrases "at least one of A, B, and C" or "at least one of A, B, or C" refers to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0045] It is understood that the specific order or hierarchy of steps, operations, or processes disclosed is an illustration of exemplary approaches. Unless explicitly stated otherwise, it is understood that the specific order or hierarchy of steps, operations, or processes may be performed in different order. Some of the steps, operations, or processes may be performed simultaneously. The accompanying method claims, if any, present elements of the various steps, operations or processes in a sample order, and are not meant to be limited to the specific order or hierarchy presented. These may be performed in serial, linearly, in parallel or in different order. It should be understood that the described instructions, operations, or systems can generally be integrated together in a single software/hardware product or packaged into multiple software/hardware products.

[0046] In one aspect, a term coupled or the like may refer to being directly coupled. In another aspect, a term coupled or the like may refer to being indirectly coupled.

[0047] Terms such as top, bottom, front, rear, side, horizontal, vertical, and the like refer to an arbitrary frame of reference, rather than to the ordinary gravitational frame of reference. Thus, such a term may extend upwardly, downwardly, diagonally, or horizontally in a gravitational frame of reference.

[0048] The disclosure is provided to enable any person skilled in the art to practice the various aspects described herein. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology. The disclosure provides various examples of the subject technology, and the subject technology is not limited to these examples. Various modifications to these aspects will be readily apparent to those skilled in the art, and the principles described herein may be applied to other aspects.

[0049] All structural and functional equivalents to the elements of the various aspects described throughout the disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112 (f), unless the element is expressly recited using the phrase "means for" or, in the case of a method claim, the element is recited using the phrase "step for".

[0050] Those of skill in the art would appreciate that the various illustrative blocks, modules, elements, components, methods, and algorithms described herein may be implemented as hardware, electronic hardware, computer software, or combinations thereof. To illustrate this interchangeability of hardware and software, various illustrative blocks, modules, elements, components, methods, and algorithms have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application. Various components and blocks may be arranged differently (e.g., arranged in a different order, or partitioned in a different way) all without departing from the scope of the subject technology.

[0051] The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately claimed subject matter.

[0052] The claims are not intended to be limited to the aspects described herein, but are to be accorded the full scope consistent with the language of the claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirements of the applicable patent law, nor should they be interpreted in such a way.

What is claimed is:

1. A method for level two or level three autonomy, the method comprising:

monitoring, by an automotive safety integrity level-D (ASIL-D) power management integrated circuit (PMIC), operation of a first ASIL-D safety island domain of a first system on chip (SOC), wherein the first SOC comprises the first ASIL-D safety island domain communicatively connected with one or more ASIL-B subsystems of an ASIL-B domain;

determining, by the ASIL-D PMIC, a fault condition associated with the first ASIL-D safety island domain; and

based on determining that there is the fault condition, transmitting an indication to shutdown one or more components of the first SOC to a safe state.

2. The method of claim **1**, further comprising monitoring, by a second ASIL-D safety island domain of a second SOC.

3. The method of claim **1**, wherein the monitoring operation of the first ASIL-D safety island domain comprises monitoring one or more of power supplies of the first ASIL-D safety island domain.

4. The method of claim **1**, wherein the monitoring operation of the first ASIL-D safety island domain comprises monitoring for one or more temperatures.

5. The method of claim **1**, wherein the monitoring operation of the first ASIL-D safety island domain comprises monitoring one or more watchdog timers of the first ASIL-D safety island domain.

6. The method of claim **1**, further comprising transmitting, based on the indication of the fault condition, an indication to cease operation of one or more applications or an indication to enter into a degraded mode of operation.

7. The method of claim **6**, wherein the degraded mode of operation comprises reducing from level three autonomy to level two autonomy.

8. The method of claim **6**, wherein the one or more applications comprise an autonomous driving application.

9. A system for monitoring safety in an autonomous vehicle, comprising:

an automotive safety integrity level-B (ASIL-B) domain;

an ASIL-D safety island domain communicatively connected with the ASIL-B domain; and

an ASIL-D power management integrated circuit (PMIC) communicatively connected with the ASIL-D safety island domain, wherein the ASIL-D PMIC monitors operation of the ASIL-D safety island domain and initiates a shutdown sequence upon detecting a fault condition.

10. The system of claim **9**, further comprising:

an ASIL-B PMIC communicatively connected with the ASIL-B domain, wherein the ASIL-B PMIC comprises safety mechanisms achieving ASIL-B integrity across voltage rails powering the ASIL-B domain.

11. The system of claim **9**, wherein the monitoring operation of the first ASIL-D safety island domain comprises monitoring one or more of power supplies of the first ASIL-D safety island domain.

12. The system of claim **9**, wherein the monitoring operation of the first ASIL-D safety island domain comprises monitoring for one or more temperatures.

13. A method for level two or level three autonomy, the method comprising:

monitoring, by an automotive safety integrity level-D (ASIL-D) power management integrated circuit (PMIC), operation of a first ASIL-D safety island domain of a first system on chip (SOC), wherein the first SOC comprises the first ASIL-D safety island domain communicatively connected with one or more ASIL-B subsystems of an ASIL-B domain;

determining, by the ASIL-D PMIC, a fault condition associated with the first ASIL-D safety island domain; and

based on determining that there is the fault condition, transmitting an indication to enter into a degraded mode of operation.

14. The method of claim **13**, further comprising monitoring, by a second ASIL-D safety island domain of a second SOC, the first ASIL-D safety island domain of the first SOC.

15. The method of claim **13**, wherein the monitoring operation of the first ASIL-D safety island domain comprises monitoring one or more power supplies of the first ASIL-D safety island.

16. The method of claim **13**, wherein the monitoring operation of the first ASIL-D safety island comprises monitoring for one or more temperatures.

17. The method of claim **13**, wherein the monitoring operation of the first ASIL-D safety island comprises monitoring one or more watchdog timers of the first ASIL-D safety island.

18. The method of claim **13**, further comprising transmitting, based on the indication of the fault condition, an indication to shutdown one or more components of the first SOC to a safe state.

19. The method of claim **18**, wherein the degraded mode of operation comprises changing from level three autonomy to level two autonomy.

20. The method of claim **18**, wherein the degraded mode of operation comprises shutdown of an autonomous driving application.

* * * * *